



Lecture

4-5

Theory of demand and supply analysis (Numerical Problems)



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- The selling price in the market = Rs. 100 per unit
- Tax per unit = Rs. 10
- How much the producer actually gets?
 $100 - 10 = \text{Rs. } 90$

So in the **supply function**, instead of using market price we use **P_t-Tax** (net price to producer)

$$S_t = g(P_t - \text{tax})$$

A subsidy is financial assistance from the Govt. that reduces the effective price paid by consumer (for demand side)

So buyer's effective price becomes: **P_t - subsidy**

So the **demand equation becomes:**

$$D_t = f(P_t - \text{subsidy})$$



Problem No: 1

The demand and supply equations of a commodity are given as $X_d = \frac{1}{2} (5 - P)$ and $X_s = 2P - 3$.

- i. Find the equilibrium price and quantity?
- ii. Find the new price and quantity if a tax of Rs. $\frac{6}{5}$ per unit is imposed on the commodity?
- iii. Find the total tax revenue generated by the government?
- iv. Find the price actually realized by the seller?

Solution

Given:

Demand equation: $X_d = \frac{1}{2} (5 - P)$

Supply equation: $X_s = 2P - 3$

i. Find the equilibrium price and quantity:

At equilibrium, $X_d = X_s$

Therefore,

$$\frac{1}{2} (5 - P) = 2P - 3$$

Now solve for P:

$$5 - P = 4P - 6$$

$$5 + 6 = 4P + P$$

$$11 = 5P$$

$$P = 11 / 5 = \text{Equilibrium price (P)}$$

Find the equilibrium quantity

Demand equation: $X_d = \frac{1}{2} (5 - P)$

$$X_d = \frac{1}{2} (5 - (11/5))$$

$$X_d = \frac{1}{2} (14/5)$$

$$X_d = 7/5 = \text{Equilibrium quantity}$$

Solution

ii. Find the new price and quantity if a tax of Rs. 6/5 per unit is imposed:

➤ When a tax is imposed on the commodity, the new supply equation becomes

$$X_s' = 2\{P' - T\} - 3 \quad (\text{where, } T \text{ is the tax per unit})$$

$$X_s' = 2\{P' - (6/5)\} - 3$$

$$X_s' = 2P' - (12/5) - 3$$

$$X_s' = 2P' - (27/5) \quad (\text{New Supply Equation})$$

At equilibrium, $X_d = X_s'$

$$\frac{1}{2}(5 - P') = 2P' - (27/5)$$

$$5 - P' = 4P' - (54/5)$$

$$3P' = 5 + (54/5)$$

$$P' = 79/25 = \text{New equilibrium price}$$

Find the equilibrium quantity

$$X_s' = 2P' - (27/5) \quad (P' = 79/25)$$

$$X_s' = 2 * (79/25) - (27/5)$$

$$X_s' = (158/25) - (27/5)$$

$$X_s' = (23/25) = \text{New equilibrium quantity}$$

Solution

- iii. Find the total tax revenue generated by the government:

Tax revenue = Tax per unit * Quantity sold with tax

$$\begin{aligned}\text{Tax revenue} &= (6/5) * X_s' \\ &= (6/5) * (23/25)\end{aligned}$$

$$\text{Tax revenue} = 138/125$$

- iv. Find the price actually realised by the seller:

Price realized by seller = New equilibrium price - Tax per unit

$$\text{Price realized by seller} = 79/25 - 6/5$$

$$\text{Price realized by seller} = 49/25$$



Problem No: 2

Solve the following problem: The demand function is $P=5-2X$ and the supply function is $P=\frac{1}{2}(X+5)$ respectively.

- i. Find the equilibrium price and quantity?
- ii. Find the new price and quantity if a subsidy of Rs. $5/2$ per unit is granted on the commodity?
- iii. Find the total amount of subsidy granted by the government?

Solution

Given:

Demand equation: $P = 5 - 2X$

Supply equation: $P = \frac{1}{2}(X + 5)$

i. Find the equilibrium quantity:

At equilibrium, $f(P) = g(P)$

$$5 - 2X = \frac{1}{2}(X + 5)$$

$$10 - 4X = X + 5$$

$$5X = 5$$

$$X = 1 = \text{Equilibrium quantity}$$

Find the equilibrium price

Demand equation: $P = 5 - 2X$

$$P = 5 - 2(1)$$

$$\text{Equilibrium price} = P = 3$$

ii. Find the new price and quantity if a subsidy of Rs. $5/2$ per unit is granted

When a subsidy is granted on the commodity, the new demand equation becomes:

$$\{P' - S\} = 5 - 2X' \quad (\text{where } S \text{ is the subsidy per unit})$$

$$\{P' - (5/2)\} = 5 - 2X' \quad S = 5/2$$

$$(2P' - 5)/2 = 5 - 2X'$$

$$(2P' - 5) = 10 - 4X'$$

$$P' = (15 - 4X')/2$$

At equilibrium, $f(P) = g(P)$

$$(15 - 4X')/2 = \frac{1}{2}(X' + 5)$$

$$15 - 4X' = X' + 5$$

$$5X' = 10$$

$$X' = 2 = \text{Equilibrium quantity}$$

Solution

Substitute the new equilibrium quantity back into either the demand or supply equation to find the new equilibrium price (P').

Let's use the supply equation:

$$P' = \frac{1}{2}(X' + 5)$$

$$P' = \frac{1}{2}(2+5)$$

$$P' = 7/2$$

So, with the subsidy granted, the new price is approximately $7/2$ and the new quantity is 2.

iii. Find the total amount of subsidy granted by the government?

Total subsidy = New quantity demanded * subsidy per unit

$$= 2 * (5/2)$$

Total subsidy = 5

Problem No: 3

- i. Suppose the value of demand and supply curves of a Commodity-X is given by the following two equations simultaneously:

$$Q_d = 200 - 10P;$$

$$Q_s = 50 + 15P$$

Find the equilibrium price and equilibrium quantity of commodity X.

- ii. Suppose that the price of a factor inputs used in producing the commodity has changed, resulting in the new supply curve given by the equation

$$Q_s' = 100 + 15P'$$

Analyze the new equilibrium price and new equilibrium quantity as against the original equilibrium price and equilibrium quantity.

Solution

Given:

Demand equation: $Q_d = 200 - 10P$

Supply equation: $Q_s = 50 + 15P$

i. Find the equilibrium price and quantity:

$$Q_d = Q_s$$

$$200 - 10P = 50 + 15P$$

$$150 = 25P$$

$$P = 6 = \text{Equilibrium price}$$

Find the equilibrium quantity

$$Q_d = 200 - 10P$$

$$Q_d = 200 - (10)(6) = 140 \text{ units}$$

ii. If the price of factor of production has changed, then under the new conditions

Given;

New supply chain equation;

$$Q_d = Q_s'$$

$$200 - 10P' = 100 + 15P'$$

$$25P' = 100$$

$$\text{Equilibrium Price } P' = 4$$

$$Q_s = 200 - (10)(4) = 160 \text{ units}$$

Thus as the equilibrium price is decreasing the equilibrium quantity is increased.



Problem No: 4

The demand function = $3Q + 4P = 24$ and the supply function is $P = \frac{1}{4}Q + 3$ respectively.

- i. Find the equilibrium price and quantity?
- ii. Find the new price and quantity if a tax of Rs. $\frac{1}{3}$ per unit is imposed on the commodity?
- iii. Find the total tax revenue generated by the government?
- iv. Find the price actually realised by the seller?

Solution

Given:

Demand equation: $D_x = 3Q + 4P = 24$

$$4P = 24 - 3Q$$

$$P = 6 - (3/4)Q \dots\dots\dots(1)$$

Supply equation: $P = \frac{1}{4}Q + 3 \dots\dots\dots(2)$

i. Find the equilibrium price and quantity:

At equilibrium, $D_x = S_x$

$$6 - (3/4)Q = (1/4)Q + 3$$

$$Q = 3 = \text{quantity demanded}$$

$$P = (1/4)Q + 3$$

$$P = \frac{1}{4}(3) + 3$$

$$P = \frac{3}{4} + 3$$

$$P = 15/4 = \text{Equilibrium price (P)}$$

ii. Find the new price and quantity if a tax of Rs. 1/3 per unit is imposed:

➤ When a tax is imposed on the commodity, the new supply equation becomes

$$\{P' - T\} = \frac{1}{4}Q + 3 \text{ (where, T is the tax per unit)}$$

$$P' - 1/3 = \frac{1}{4}Q + 3$$

$$P' = \frac{1}{4}Q + 10/3 \dots\dots\dots(3) \text{ (New Supply Equation)}$$

At equilibrium,

$$\frac{1}{4}Q + 10/3 = 6 - (3/4)Q \quad (\text{Eq3} = \text{Eq1})$$

$$Q' = 6 - (10/3) = 8/3$$

$$P' = \frac{1}{4}Q + 10/3$$

$$P' = \frac{1}{4}(8/3) + 10/3 = 8/12 + 10/3$$

$$P' = 4$$

Solution

iii. Find the total tax revenue generated by the government:

Tax revenue = Tax per unit * Quantity sold with tax

$$\begin{aligned}\text{Tax revenue} &= (1/3) * X_s' \\ &= (1/3) * (8/3)\end{aligned}$$

$$\text{Tax revenue} = 8/9$$

iv. Find the price actually realised by the seller:

Price realized by seller = New equilibrium price - Tax per unit

$$\text{Price realized by seller} = 4 - 1/3$$

$$\text{Price realized by seller} = 11/3$$